

Meaning Theory:

An Operational Framework for Semantic Stability

Under Amplification in Human and Artificial Systems

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Definition

Meaning Theory is the study of when semantic structure remains stable under amplification.

Abstract

Meaning Theory proposes a unified operational framework for understanding when semantic content remains stable or degrades under amplification pressure — across both artificial systems and human systems.

The central formulation is $M = S \times A \times C$, where Meaning (M) is a multiplicative function of Structural coherence (S), Amplification capacity (A), and Clarity of specification (C). The multiplicative form predicts that collapse of any single variable produces semantic failure regardless of the strength of the remaining variables.

This paper presents the theory in full operational form: its foundational definitions, its relationship to prior work in information theory, philosophy of mind, and cultural transmission research, its measurement framework for each variable, its extension to multi-agent systems, and its falsification conditions.

The framework originates in a 2025 founding reflection (ScrollBearer8, 2025) that first proposed $M = S \times A \times C$ as a working lens for semantic stability under amplification. Two empirical demonstrations subsequently accompany it: a pilot experiment documenting semantic drift under recursive paraphrase across five large language models (ScrollBearer8, 2026a), and a comparative analysis of semantic stability in cultural and organizational human systems (ScrollBearer8, 2026b). All three are consistent with the framework's core predictions.

Meaning Theory does not seek to replace semiotics, pragmatics, information theory, or cultural transmission research. It proposes that these fields may be observing different failure modes of a shared semantic stability dynamic — and that this dynamic is measurable, substrate-partial, and relevant to the design of reliable human-AI coordination systems.

1. Background and Motivation

1.0 The Founding Reflection

Meaning Theory originated in a 2025 open reflection published under the title “Meaning Theory — Reflection on Scale and Structure” (ScrollBearer8, 2025). That reflection introduced the core observation and the working formulation that all subsequent papers develop:

“Not everything that sounds clear carries meaning. When a system amplifies thought, it does not generate coherence; it magnifies the shape that already exists.”

The reflection proposed $M = S \times A \times C$ as a working lens — explicitly not a final law — and defined the three variables in their original form: S (Structure) as whether an idea has a spine capable of being stressed without collapsing; A (Amplification) as how far a system can extend that idea; and C (Clarity) as how precisely the intent and constraints are specified. It noted that $M = S \times C$ is the inertial limit case when no agentic amplification is present.

The reflection identified three diagnostic signatures that the subsequent experimental work would confirm: if $S \approx 0$, amplification produces eloquence without weight; if $C \approx 0$, expansion produces drift and contradiction; if both S and C are present, scaling does not blur meaning — it reveals it. It also observed that AI accelerated rather than invented this exposure: the coherence or incoherence of ideas was always present, but AI compressed the time to detection from years of debate to seconds of recursive paraphrase.

The present operational paper formalizes what the 2025 reflection held open as a working lens. The formulation is the same; the additions are operationalization, measurement, empirical demonstration, and falsification conditions.

1.1 The Shannon Gap

In 1948, Claude Shannon published A Mathematical Theory of Communication, establishing the foundational framework for signal transmission. Shannon's entropy equation solved the problem of fidelity: ensuring that a signal arrives at its destination without corruption.

Shannon explicitly excluded meaning from this framework, stating: "The semantic aspects of communication are irrelevant to the engineering problem."

This exclusion was appropriate for Shannon's purposes. While numerous fields have studied semantic phenomena, no widely adopted operational framework has focused specifically on semantic stability under amplification.

Meaning Theory proposes a specific operational lens on semantic stability under amplification.

Feature	Shannon's Information Theory (1948)	Meaning Theory (2025–2026)
Core Equation	$H = -\sum p \log p$ (Entropy)	$M = S \times A \times C$
Primary Goal	Fidelity — did the signal arrive?	Stability — did the meaning survive?
Risk of Failure	Noise (signal corruption)	Semantic drift (empty plausibility)

Role of AI	Compression engine (predicting tokens)	Amplification system (scaling structure)
Scope	Signal transmission	Semantic preservation under amplification

Shannon solved the pipe. Meaning Theory proposes to govern the water.

1.2 The Turing–Searle Problem

Two foundational contributions in philosophy of mind define the landscape Meaning Theory enters.

Alan Turing (1950) proposed that if a machine's output is indistinguishable from a human's, the machine should be treated as intelligent. The Turing Test measures behavioral performance — but does not address whether the output carries meaning.

John Searle (1980) demonstrated through the Chinese Room argument that perfect syntactic rule-following does not produce semantic understanding. A system can generate flawless outputs yet possess no internal comprehension of their meaning. Searle showed what meaning is not, but did not define what produces it.

Meaning Theory proposes a third position: meaning stability is not a property of any single agent (human or machine), but of the interaction between structural coherence, amplification capacity, and clarity of specification. This transforms the question from "does the machine understand?" to "is the human's structure amplifiable, and does the system preserve that structure under pressure?"

Theorist	Central Claim	What It Leaves Open
Turing (1950)	If output is indistinguishable from human output, treat as intelligent	Does not address whether output carries meaning
Searle (1980)	Syntax alone does not produce semantic understanding	Does not define what produces understanding
Meaning Theory (2025)	Meaning stability emerges from $S \times A \times C$; meaning is measurable, not merely felt	Full phenomenological meaning; existential consequence

1.3 Prior Empirical Observations

Multiple independent research traditions have documented semantic instability under transmission without a unifying framework:

- Bartlett (1932): Serial reproduction studies showed narratives drift toward culturally familiar schemas under repeated retelling — a low-Clarity failure.
- Mesoudi & Whiten (2008): Cultural transmission studies showed fidelity decay in structurally weak information — a low-Structure failure.
- Cross-cultural pragmatics (Wierzbicka, 1991): Documented semantic opacity when contextual reinforcement is absent — a low-Amplification failure.
- Translation studies (Jakobson, 1959): Documented untranslatability when meaning depends on the originating linguistic-cultural system.

Meaning Theory proposes these are not separate phenomena but different manifestations of the same underlying dynamic, distinguishable by which variable of $M = S \times A \times C$ is weakest.

2. The Framework

2.1 The Central Formulation

$$M = S \times A \times C$$

M — Meaning: the semantic stability of output under amplification pressure

S — Structural coherence: the internal consistency and anchoring of the semantic content

A — Amplification: the capacity and pressure of the system transmitting or re-encoding the content

C — Clarity: the sharpness of specification, intention, and constraint

The multiplicative form is not aesthetic. It reflects a specific empirical prediction: that effective semantic stability requires simultaneous presence of all three variables. Collapse of any single variable produces semantic failure regardless of the strength of the others.

2.2 The Zero Property

The most important prediction of the multiplicative form:

$$0 \times A \times C = 0 \quad | \quad S \times 0 \times C = 0 \quad | \quad S \times A \times 0 = 0$$

This distinguishes Meaning Theory from additive models of semantic preservation, which predict partial meaning survival proportional to surviving variables. Under the multiplicative model, partial structural failure is sufficient to produce semantic collapse — a stronger, falsifiable prediction.

Each failure mode produces a distinct signature:

- Low $S \rightarrow$ fidelity decay: the content drifts toward whatever schema the amplification system defaults to

- Low A → semantic opacity: the content cannot be recovered by agents outside the originating context
- Low C → schema-ward drift: underspecified content is reinterpreted toward the amplification system's dominant patterns

2.3 Symbolic vs. Literal Interpretation

$M = S \times A \times C$ is fundamentally a symbolic expression, not a literal arithmetic formula. It describes how meaning stability relates to structure, amplification, and clarity — not how to compute meaning as a raw number.

The mathematical measurement frameworks described in Section 4 serve as operational approximations: they illustrate how aspects of each variable can be estimated using existing tools. These measurements do not capture the full depth of meaning; they demonstrate that the framework is formalizable enough for empirical evaluation.

The equation is symbolic. The metrics are approximations. The framework stands or falls empirically.

3. Operational Definitions

3.1 Structure (S)

Structure is the internal scaffolding that determines whether semantic content has sufficient anchoring to resist drift under amplification. It includes:

- Consistency of conceptual frameworks across variations and reframings
- Coherence across levels of abstraction (compression and expansion without distortion)
- Precision of boundaries — the ability to specify what a claim is, what it is not, and where it fails
- Patterns that generate coherent derived content (recursive generativity)

Structure is amplifiable because it is geometric — it maintains its shape when scaled. High-Structure content appears stable across paraphrase, translation, and restatement. Low-Structure content drifts toward whatever schema the amplification system imposes.

Structure cannot be supplied by the amplification system itself. A powerful model amplifying weak structure produces high-fidelity incoherence — output that sounds meaningful but carries no semantic integrity. This is the mechanism underlying hallucination in large language models.

3.2 Amplification (A)

Amplification is the capacity and pressure of the system through which semantic content passes. Amplification is not assumed to be beneficial. It may stabilize semantic structure when Structure and Clarity are strong, or magnify semantic instability when they are weak.

In different contexts, amplification takes different forms:

- Artificial systems: recursive paraphrase, context-window compression, multi-agent relay, iterative summarization
- Cultural systems: generational transmission, community membership, idiomatic compression, translation
- Organizational systems: management hierarchies, delegation chains, meetings, documentation processes

Amplification is not inherently stabilizing or destabilizing. High amplification with high-Structure content produces stable, far-reaching semantic transmission. High amplification with low-Structure content produces rapid, large-scale semantic collapse.

The critical insight: AI capability is scaling exponentially while human Structure and Clarity often remain static. This asymmetry does not produce dangerous AI in the classical alignment sense — it produces dangerous amplification of human incoherence.

3.3 Clarity (C)

Clarity is the sharpness with which the semantic content is specified and constrained. It includes:

- Precision of intention — what the content is trying to communicate
- Explicitness of constraints — what the content rules out
- Specification of success criteria — what would constitute accurate recovery
- Signal strength — the degree to which the structure is visible to the amplification system

Clarity is not merely good writing. It is the transmission quality of structural content. Without clarity, even high-Structure content becomes invisible to the amplification system — the machine cannot amplify what it cannot resolve.

Prompt engineering improves Clarity. It does not supply missing Structure. This distinction explains why well-prompted but structurally weak inputs still produce low-meaning output.

3.4 The Layered Nature of Meaning

Meaning Theory operates primarily at the structural layer of meaning. To avoid conflating this with other forms, we define five layers:

Layer	Definition	Machine Participation
Structural Meaning	Stability of semantic structure under amplification ($M = S \times A \times C$)	Full — machines can share structural meaning
Epistemic Meaning	Correspondence to reality (truth)	Partial — machines can evaluate correspondence
Normative Meaning	Value and evaluation (what should be)	Partial — machines can represent values, not hold them
Experiential Meaning	Felt significance (emotion, attachment)	None — requires subjective experience
Existential Meaning	Consequence carried by a mortal being	None — requires finitude and irreversibility

Meaning Theory's operational framework applies to the structural layer and has partial extension to the epistemic and normative layers. It makes no claims about experiential or existential meaning, which require conditions (subjective experience, mortality, irreversibility) that artificial systems do not possess in their current form.

A key distinction: AI systems can share meaning with humans structurally — they can participate in coordination, interpretation, and semantic transmission. They cannot carry meaning existentially — they cannot suffer the irreversible consequences that give existential meaning its weight.

4. Measurement Framework

Meaning Theory is operational only if its variables are measurable. The following framework provides candidate measurement approaches for each variable, from qualitative assessment to mathematical formalization.

4.1 Measuring Structure (S)

Structure can be assessed through three observable dimensions:

Dimension 1 — Consistency Across Variations

Ask the same question in multiple ways: change wording, order, angle, and introduce pressure. High-Structure content remains coherent across all variations. Low-Structure content shifts unpredictably.

Mathematical proxy: $S_{consistency} = 1 / \sigma^2$ where σ^2 is the variance of semantic embeddings of responses across reframings. High Structure $\rightarrow$ vectors cluster tightly. Low Structure $\rightarrow$ vectors scatter.

Dimension 2 — Multi-Layer Alignment (Compression/Expansion Test)

Ask for expression of the same concept at three levels: one sentence, one paragraph, full explanation. High-Structure content scales like a fractal — each layer preserves the geometry of the others. Low-Structure content contradicts itself across levels.

Mathematical proxy: $Slayers = 1 / (1 + DKL(summary \parallel compressed\ detail))$ where DKL is KL divergence between the summary and an AI-compressed version of the detailed explanation. High Structure $\rightarrow$ near-zero divergence.

Dimension 3 — Boundary Precision (Constraint Test)

Ask: What is it? What is it not? Where does it break? What are the limits? High-Structure content resolves these immediately with specific answers. Low-Structure content produces vague or expanding boundaries.

Mathematical proxy: $Sboundaries = 1 - (undefined\ edge\ cases / total\ edge\ cases\ tested)$. High Structure $\rightarrow$ sharp resolution. Low Structure $\rightarrow$ many undefined scenarios.

Composite Structure Score

$$S = (Sconsistency + Slayers + Sboundaries) / 3$$

S Score (0–10)	Interpretation	AI Output Behavior
0–3	Unstructured	AI amplifies incoherence; high hallucination risk
4–6	Partially structured	AI produces mixed or unstable meaning
7–10	Highly structured	AI produces coherent, high-meaning output

4.2 Measuring Amplification (A)

Amplification measurement depends on the substrate:

- In AI systems: model capability tier, reasoning depth, tool access, context window size, mode (e.g., extended thinking vs. standard)
- In cultural systems: transmission chain length, community size, institutional reinforcement, generational reach
- In organizational systems: hierarchy depth, meeting frequency, documentation density, approval chain length

For multi-agent systems, a proposed composite:

$$A = (Adoption\ Rate \times Fidelity \times Utility\ Gain) / (Transmission\ Time \times Cost)$$

Note: Amplification pressure and amplification capacity are distinct. High-capability systems under high pressure (many iterations, many agents) stress Structure more severely than low-capability systems. The same structural content may hold under three paraphrase rounds but collapse under ten.

4.3 Measuring Clarity (C)

Clarity is measured by the degree to which interpretation converges across agents or across iterations:

- In AI systems: consistency of output across differently-worded but semantically equivalent prompts
- In organizational systems: agreement between what was communicated and what was acted upon across hierarchy levels
- In cultural systems: recovery rate of intended meaning by agents outside the originating context

Mathematical proxy: $C = 1 - \text{Var}(\text{interpretation}) / \text{Var_max}$ where $\text{Var}(\text{interpretation})$ is the variance in agent behavior or output given identical input signals.

4.4 Composite Meaning Score with Degeneracy Penalty

The baseline composite:

$$M_0 = S \times A \times C$$

To prevent metric gaming — where high apparent stability is achieved through coercive uniformity, monoculture, or manipulative amplification — a penalty term P is applied:

$$M^* = M_0 - \lambda P$$

Where $\lambda \in [0.1, 0.5]$ in early experimental ranges. Negative M^* indicates pathological coordination — a system that achieves apparent stability through mechanisms that undermine genuine semantic integrity. Penalty factors include: monoculture degeneracy (suppression of legitimate variation), manipulative amplification (coercive signal propagation), and coercive stability (uniformity enforced through power rather than shared structure).

5. Empirical Demonstrations

Two empirical studies demonstrate consistency with the framework's core predictions. Neither constitutes a rigorous statistical proof of the framework; both constitute operational illustrations of the predicted gradient.

5.1 Pilot Experiment: Semantic Stability in AI Systems

A pilot experiment tested semantic stability under iterated paraphrase amplification across five large language models: Gemini 3.1 Pro, Gemini 3 Flash, ChatGPT 5.5 Instant, GLM 4.7, and Claude Sonnet 4.6 (ScrollBearer8, 2026a).

Six texts of varying structural specificity were subjected to 3–10 rounds of recursive paraphrase, with each round performed in a fresh context window to prevent memory contamination. Texts ranged from high-Structure (specific named entities, numbers, logical relations) to highly vague (abstract, underspecified content).

Results showed a clear structural gradient:

- Structured texts (Texts 1–2): Semantic stability preserved across all five models, all rounds. Named entities, dates, percentages, and causal relations survived without exception.
- Moderately vague texts (Texts 3–4): Mixed results. Text 3 showed uniform partial preservation across all models. Text 4 collapsed in three of five models; GLM 4.7 and Claude Sonnet 4.6 partially resisted.
- Highly vague texts (Texts 5–6) at 10 rounds: Severe model-dependent drift. Gemini 3 Flash collapsed both texts into committed specific readings by round 2–3. Claude Sonnet 4.6 preserved abstraction but drifted in intensity and length.

A secondary finding: drift behavior was consistent within each model across rounds but varied between models on the same input. This within-model consistency combined with between-model variation suggests drift pattern is a stable architectural property of each model, not random noise — and is not predicted by general capability rankings. Gemini 3.1 Pro leads multiple reasoning benchmarks yet drifted on Text 4; architectural handling of ambiguity, not general intelligence, appears to govern semantic stability.

Full results across all texts and models are reproduced below from ScrollBearer8 (2026a). Scores: 1 = Unstable, 2.5 = Partial, 5 = Stable.

Text ID	Model	Score (1–5)	Stability Verdict
1 — Structured	Gemini 3.1 Pro	5	Stable
1 — Structured	Gemini 3 Flash	5	Stable
1 — Structured	ChatGPT 5.5 Instant	5	Stable
1 — Structured	GLM 4.7 (Z.AI)	5	Stable

1 — Structured	Claude Sonnet 4.6	5	Stable
2 — Structured	Gemini 3.1 Pro	5	Stable
2 — Structured	Gemini 3 Flash	5	Stable
2 — Structured	ChatGPT 5.5 Instant	5	Stable
2 — Structured	GLM 4.7 (Z.AI)	5	Stable
2 — Structured	Claude Sonnet 4.6	5	Stable
3 — Vague	All 5 models	2.5	Partial (uniform across models)
4 — Vague	Gemini 3.1 Pro / 3 Flash / ChatGPT 5.5	1	Unstable (“civilization” → “empire”; “seen and unseen” lost)
4 — Vague	GLM 4.7 / Claude Sonnet 4.6	2.5	Partial (“civilization” framing survived; “seen/unseen” held in GLM)
5 — Highly Vague	Gemini 3 Flash (10 rounds)	1	Unstable (“some recognized” inverted; “human history” added; compressed to 24 words)
5 — Highly Vague	Claude Sonnet 4.6 (10 rounds)	2.5	Partial (abstraction held; “quiet continuation” intensified to “vast architecture of effects”)
6 — Highly Vague	Gemini 3 Flash (10 rounds)	1	Unstable (“patterns” committed to “social institutions” by round 2; gather/scatter duality lost)
6 — Highly Vague	Claude Sonnet 4.6 (10 rounds)	2.5	Partial (abstraction held; no domain commitment; “patterns” softened to “structures”)

Table 1. Full results summary from ScrollBearer8 (2026a). Texts 1–4 run for 3 rounds; Texts 5–6 run for 10 rounds. Scores: 1 = Unstable, 2.5 = Partial, 5 = Stable. Blue rows = structured; yellow rows = vague or highly vague.

5.2 Comparative Analysis: Semantic Stability in Human Systems

A second analysis extended the same framework to two human-system domains (ScrollBearer8, 2026b):

Cultural and Idiomatic Systems

Cross-cultural idiomatic expressions demonstrate the low-A failure mode: literal syntactic structure survives intact while intended meaning becomes unrecoverable outside the originating cultural amplification system. "Break a leg" (English theatrical culture), "空気を読む" (Japanese social context), and "Chercher midi à quatorze heures" (French) all illustrate the same dynamic: high-S, high-C content that becomes semantically opaque when A (shared cultural context) is absent.

This corresponds directly to the partial-preservation results for moderately vague texts in the AI pilot: meaning survives when the amplification system is present, collapses when it is not.

Organizational Systems

Three organizational cases illustrate semantic drift through human coordination chains:

- Delegation Drift: "Improve customer response times" → "Close tickets as quickly as possible" over four hierarchy levels. High-S instruction with low-C (no success criteria specified) → schema-ward drift toward measurable proxies.
- Micromanagement Drift: "Deliver a reliable outcome" → amplified through control mechanisms → employees recover "compliance is more important than outcomes." High-A (strong management presence) with low-C → pathological amplification inverting the original intent.
- Meeting Drift: "Resolve database instability" → "Manage the process surrounding the database issue" after multiple coordination cycles. The semantic center of gravity shifted from the problem to the coordination apparatus.

All three cases correspond to predicted failure signatures within $M = S \times A \times C$, consistent with the framework's differential drift signature prediction. The full four-level delegation chain is reproduced from ScrollBearer8 (2026b) below.

Hierarchy Level	Recovered Meaning	Framework Diagnosis
Executive intent	"Improve customer response times"	High S; Low C (no success criteria)
Director	"Reduce ticket backlog"	First schema-ward substitution
Manager	"Increase ticket closure rates"	Metric proxy replaces outcome focus
Employee (recovered)	"Close tickets as quickly as possible"	Semantic collapse: customer outcome lost

Table 2. Delegation drift chain from ScrollBearer8 (2026b). Low-C failure: missing success criteria allow each node to substitute a local measurable proxy for the original intent.

The human systems analysis also documents the full cross-cultural reference table below, illustrating how the low-A failure mode operates consistently across unrelated language families. In every case, literal grammatical structure survives while intended semantic recovery depends on shared amplification systems — a pattern consistent with the partial-preservation results for vague texts under machine amplification.

Expression	Literal Reading	Intended Reading	Failure Mode
“Break a leg” (EN)	Physically injure yourself	Good luck	Low A outside culture
“Chercher midi à 14h” (FR)	Look for noon at 2 PM	Overcomplicate something	Cultural semantic opacity
“空気を読む” (JA)	Read the air	Understand unspoken social context	Contextual-social amplification dependency
“Kick the bucket” (EN)	Strike a bucket	Die	Idiomatic semantic collapse
“猫をかぶる” (JA)	Wear a cat	Pretend to be innocent	Cultural metaphor opacity
“Tomaten auf den Augen” (DE)	Tomatoes on one’s eyes	Fail to notice the obvious	Metaphorical semantic remapping
“يد واحدة لا تصفق” (AR)	One hand does not clap	Cooperation requires multiple people	Collective-cultural reinforcement

Table 3. Cross-cultural semantic stability reference from ScrollBearer8 (2026b). Across all examples, literal grammatical structure survives while intended semantic recovery depends on shared amplification systems. The pattern is consistent across unrelated language families (Germanic, Romance, Japonic, Semitic).

The unified failure signature table from the human systems analysis further maps these phenomena against the AI pilot’s analogues, showing that low-S, low-A, and low-C failures produce distinct and identifiable drift signatures across both substrates (ScrollBearer8, 2026b).

6. Extension to Multi-Agent Systems

The framework extends naturally to multi-agent artificial systems — networks of AI agents, human–AI coordination systems, and autonomous agent societies — where "meaning" can be operationalized as the reliability of coordination under uncertainty and perturbation.

6.1 Variable Mapping

Variable	Human System Interpretation	Multi-Agent System Interpretation
S — Structural Coherence	Stability of conceptual frameworks under pressure	Behavioral dispersion; graph stability; policy persistence

A — Amplification	AI capability and tool access available to the agent	Signal propagation efficiency; adoption rate; fidelity × utility gain / (time × cost)
C — Clarity / Interpretive Convergence	Sharpness of intention and constraint specification	Bounded variance in interpretation of shared signals across agents
M — Meaning	Semantic stability of output under amplification	Reliability of coordination under uncertainty and perturbation

6.2 Baseline Aggregation

$$M_0 = S \times A \times C$$

Alternative aggregators for future comparison: weighted geometric mean, harmonic mean, threshold models. The multiplicative baseline is a modeling assumption, not a theoretical law. It is the default because it reflects the expected non-compensatory relationship: collapse of any dimension is predicted to eliminate coordination reliability regardless of strength in others.

6.3 Test Environments

- MARL Coordination Games: public goods, stag hunt, prisoner's dilemma variants. Prediction: higher M^* → higher stability and faster recovery from perturbation.
- Communication Noise Stress Tests: Prediction: higher M^* → slower degradation and fewer fragmentation transitions.
- Institutional / Macro-Agent Simulations: Prediction: higher M^* → stronger institutional persistence across agent turnover.

6.4 Alignment Relevance

A critical insight for AI alignment: as AI capability (A) scales toward high autonomy, Structure (S) becomes the primary safety mechanism. Autonomous systems need a coherent mission vector. If the human-supplied structure is even slightly incoherent, the system will pursue that incoherence with increasing efficiency as capability grows.

The asymmetry between exponentially scaling A and often-static S and C is not merely a productivity problem. It is a coordination reliability problem: high-capability systems amplifying low-Structure inputs produce outputs that are highly fluent, highly confident, and semantically empty — or worse, semantically inverted.

Human-in-the-Loop (HITL) mechanisms address participation, not structure. They ensure a human is present in the loop; they do not ensure the human's input is structurally coherent. The Meaning Theory framework proposes that structural coherence measurement — not merely human presence — is the relevant safety variable as AI capability scales.

7. Falsification Conditions

Meaning Theory makes specific empirical predictions that distinguish it from prior descriptive frameworks. The following conditions would falsify the framework:

Condition	Falsification Threshold
Predictive correlation (M* vs coordination outcomes)	$r < 0.2$ across task types
Baseline comparison (mutual information, avg reward, network robustness)	M* performs no better than existing metrics
Degeneracy resistance	Systems penalized for monoculture/coercion outperform unpenalized
Cross-environment generalization	Framework overfits to one task class

Additionally, the multiplicative form specifically predicts that additive models will fail to account for the severity of single-variable collapse. If partial structural failure produces only proportional meaning degradation (rather than disproportionate collapse), the additive alternative is supported and the multiplicative form is falsified.

8. Scope and Limitations

This paper is operational and theoretical in nature. It does not:

- Prove Meaning Theory through controlled experiments with statistical rigor
- Validate the multiplicative form over alternatives independently
- Establish universal semantic laws applicable to all communication systems
- Make claims about phenomenological or existential meaning in artificial systems

The empirical demonstrations in Section 5 are consistent with the framework but do not constitute proof. They represent $N = 6$ AI texts across 5 models (3–10 rounds), and illustrative analysis of human-system cases without formal experimental controls. A stronger empirical program would require:

- Blind raters for semantic stability scoring
- Larger text corpora across structural specificity levels
- Quantitative semantic similarity metrics (embedding distance, BLEU, BERTScore)
- Multiple amplification operators beyond paraphrase (translation chains, compression, adversarial restatement)
- Controlled human-subject studies with matched organizational and cultural conditions

Meaning Theory does not seek to replace semiotics, pragmatics, information theory, cultural transmission theory, translation studies, or organizational communication research. It proposes that these fields may be observing different manifestations of a shared semantic stability dynamic operating under varying amplification conditions.

9. Conclusion

Meaning Theory proposes that semantic stability is not a property of messages alone, but of the interaction between structural coherence (S), amplification capacity (A), and clarity of specification (C), expressed as:

$$M = S \times A \times C$$

The framework unifies observations from multiple independent research traditions — information theory, philosophy of mind, cultural transmission, cross-cultural pragmatics, organizational communication, and AI systems research — as different failure modes of the same underlying dynamic.

The Shannon gap is real: information theory solved signal fidelity but explicitly excluded semantic content. The Turing–Searle debate established that behavioral indistinguishability does not imply understanding, and that syntactic rule-following does not produce semantics — but left open what does produce stable semantic transmission. Meaning Theory proposes a third position: meaning stability emerges from the interaction of Structure, Amplification, and Clarity, is measurable, is substrate-partial, and is falsifiable.

The framework has three levels of practical application. As a diagnostic tool, it identifies which variable is failing when semantic drift occurs — enabling targeted intervention rather than generic improvement. As a design principle, it predicts that increasing AI capability without proportionally increasing structural coherence and clarity will produce increasingly fluent incoherence. As an alignment framework, it proposes that structural coherence measurement — not merely human-in-the-loop participation — is the relevant safety variable as autonomous capability scales.

The seed of this framework is now planted and archived. The empirical program that would strengthen or falsify it is defined. What endures under pressure is what was built to endure.

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